Computer Networks MCQ Question Bank

60 questions | Difficulty 0.5 to 10 (step 0.5, 3 questions per level) | 4 options each

Q1. [Difficulty: 0.5] What does IP stand for in 'IP address'?

A) Interconnect Port
B) Internet Protocol
C) Internet Process
D) Internal Protocol

Answer: B

Q2. [Difficulty: 0.5] What does LAN stand for?

A) Linked Area Node
B) Large Area Network
C) Local Access Network
D) Local Area Network

Answer: D

Q3. [Difficulty: 0.5] Which protocol is used to browse web pages?

A) HTTP
B) SSH
C) FTP
D) SMTP

Answer: A

Q4. [Difficulty: 1] Which protocol is used to send email?

A) SMTP
B) HTTP
C) DNS
D) FTP

Answer: A

Q5. [Difficulty: 1] Which device connects different networks and forwards packets between them?

A) Router
B) Hub
C) Network cable
D) Repeater

Answer: A

Q6. [Difficulty: 1] What does DNS do?

A) Assigns IP addresses to devices
B) Encrypts web traffic
C) Blocks malicious websites
D) Translates domain names into IP addresses

Answer: D

Q7. [Difficulty: 1.5] How many layers does the OSI model have?

A) 5
B) 7
C) 6
D) 4

Answer: B

Q8. [Difficulty: 1.5] At which OSI layer does the IP protocol operate?

A) Session layer
B) Data link layer
C) Network layer
D) Transport layer

Answer: C

Q9. [Difficulty: 1.5] Which of the following is a valid IPv4 address?

A) 256.1.1.1
B) 192.168.1.1.1
C) 192.168.1.1
D) 192.168.1

Answer: C

Q10. [Difficulty: 2] What is the default port of HTTP?

A) 80
B) 443
C) 21
D) 25

Answer: A

Q11. [Difficulty: 2] Which of the following protocols is connection-oriented?

A) IP
B) TCP
C) UDP
D) ICMP

Answer: B

Q12. [Difficulty: 2] What does URL stand for?

A) Universal Reference Link
B) Uniform Resource Locator
C) Unified Resource Location
D) Uniform Retrieval Locator

Answer: B

Q13. [Difficulty: 2.5] What is the default port of HTTPS?

A) 22
B) 80
C) 8080
D) 443

Answer: D

Q14. [Difficulty: 2.5] How many bits are there in a MAC address?

A) 32
B) 128
C) 48
D) 64

Answer: C

Q15. [Difficulty: 2.5] How many bits are there in an IPv4 address?

A) 32
B) 128
C) 64
D) 16

Answer: A

Q16. [Difficulty: 3] Which device operates at the data link layer (Layer 2)?

A) Repeater
B) Modem
C) Hub
D) Switch

Answer: D

Q17. [Difficulty: 3] Which protocol does the ping command use?

A) UDP
B) ARP
C) ICMP
D) TCP

Answer: C

Q18. [Difficulty: 3] Which of the following is a private IP address?

A) 172.32.5.1
B) 11.0.0.1
C) 8.8.8.8
D) 192.168.1.10

Answer: D

Q19. [Difficulty: 3.5] Which transport protocol is typically used for live video streaming and online games, where speed matters more than guaranteed delivery?

A) UDP
B) TCP
C) FTP
D) ICMP

Answer: A

Q20. [Difficulty: 3.5] What is the purpose of ARP?

A) To map an IP address to a MAC address
B) To assign IP addresses automatically
C) To encrypt data at the link layer
D) To map a domain name to an IP address

Answer: A

Q21. [Difficulty: 3.5] What does DHCP do?

A) Routes packets between networks
B) Translates domain names to IP addresses
C) Automatically assigns IP addresses to devices on a network
D) Encrypts data between hosts

Answer: C

Q22. [Difficulty: 4] What is the correct order of the TCP three-way handshake?

A) ACK, SYN, FIN
B) SYN, ACK, SYN-ACK
C) SYN, FIN, ACK
D) SYN, SYN-ACK, ACK

Answer: D

Q23. [Difficulty: 4] What is the main difference between HTTP and HTTPS?

A) HTTPS uses UDP instead of TCP
B) HTTPS does not use port numbers
C) HTTPS encrypts the communication using TLS
D) HTTPS only works on local networks

Answer: C

Q24. [Difficulty: 4] What is a subnet mask used for?

A) To find the DNS server
B) To assign port numbers
C) To separate the network part and the host part of an IP address
D) To encrypt the IP address

Answer: C

Q25. [Difficulty: 4.5] Which port does DNS use by default?

A) 53
B) 143
C) 67
D) 110

Answer: A

Q26. [Difficulty: 4.5] Which HTTP status code means 'Not Found'?

A) 301
B) 404
C) 500
D) 403

Answer: B

Q27. [Difficulty: 4.5] What does NAT do?

A) Translates domain names to IP addresses
B) Translates private IP addresses to a public IP address
C) Assigns MAC addresses
D) Encrypts packets at the network layer

Answer: B

Q28. [Difficulty: 5] Which HTTP method is NOT idempotent by definition?

A) PUT
B) DELETE
C) GET
D) POST

Answer: D

Q29. [Difficulty: 5] How does a switch differ from a hub?

A) A switch broadcasts every frame to all ports
B) A switch forwards frames only to the destination port using a MAC table; a hub sends them to all ports
C) A hub uses MAC addresses to filter traffic
D) A switch works only at the physical layer

Answer: B

Q30. [Difficulty: 5] What is the purpose of the TTL field in an IP packet?

A) It encrypts the packet payload
B) It sets how long a web page stays cached
C) It stores the packet's priority
D) It limits the packet's lifetime (in hops) to prevent endless looping

Answer: D

Q31. [Difficulty: 5.5] Which layer provides end-to-end delivery between applications and uses port numbers?

A) Data link layer
B) Transport layer
C) Session layer
D) Network layer

Answer: B

Q32. [Difficulty: 5.5] When you type a URL for the first time (nothing cached), what happens first?

A) The browser renders the page
B) The server sends the HTML page
C) The TLS handshake
D) A DNS lookup to find the server's IP address

Answer: D

Q33. [Difficulty: 5.5] What is the difference between HTTP status codes 301 and 302?

A) 301 is a permanent redirect; 302 is a temporary redirect
B) They are identical
C) 301 is a client error; 302 is a server error
D) 301 means success; 302 means failure

Answer: A

Q34. [Difficulty: 6] How many usable host addresses are there in a /24 IPv4 network?

A) 254
B) 253
C) 255
D) 256

Answer: A

Q35. [Difficulty: 6] Which TCP flag is used to close a connection gracefully?

A) SYN
B) PSH
C) FIN
D) RST

Answer: C

Q36. [Difficulty: 6] What is the role of a port number?

A) To identify the network the host belongs to
B) To identify the router's location
C) To identify which application or process on a host should receive the data
D) To identify the physical network card

Answer: C

Q37. [Difficulty: 6.5] Which mechanism does TCP use for flow control?

A) The slow-start threshold only
B) A sliding window using the receiver's advertised window size
C) The IP checksum
D) The three-way handshake

Answer: B

Q38. [Difficulty: 6.5] What is the difference between flow control and congestion control?

A) Congestion control is done only by the receiver
B) Flow control works only in UDP
C) Flow control stops the sender from overwhelming the receiver; congestion control stops it from overwhelming the network
D) They are two names for the same thing

Answer: C

Q39. [Difficulty: 6.5] What is the difference between a MAC address and an IP address?

A) They are both assigned by DHCP
B) An IP address is burned into the network card
C) A MAC address is a link-layer hardware address; an IP address is a logical address that depends on the network
D) A MAC address can be changed only by the ISP

Answer: C

Q40. [Difficulty: 7] How does traceroute discover the routers on a path?

A) It asks the DNS server for the route
B) It sends packets with increasing TTL values and reads the ICMP Time Exceeded replies
C) It broadcasts ARP requests to every router
D) It reads the routing table of the destination host

Answer: B

Q41. [Difficulty: 7] What is the difference between HTTP status codes 401 and 403?

A) They have the same meaning
B) 401 means not authenticated; 403 means authenticated but not allowed
C) 401 is a server error; 403 is a redirect
D) 401 means forbidden; 403 means unauthorized login is missing

Answer: B

Q42. [Difficulty: 7] What is the size of the UDP header?

A) 16 bytes
B) 4 bytes
C) 20 bytes
D) 8 bytes

Answer: D

Q43. [Difficulty: 7.5] How many usable host addresses are there in a /20 IPv4 network?

A) 1022
B) 4094
C) 4096
D) 2046

Answer: B

Q44. [Difficulty: 7.5] Which DNS record type makes one domain name an alias of another?

A) MX
B) A
C) CNAME
D) PTR

Answer: C

Q45. [Difficulty: 7.5] What does the 'Host' header in HTTP/1.1 make possible?

A) Choosing between TCP and UDP
B) Encrypting the request body
C) Hosting multiple websites on a single IP address (virtual hosting)
D) Caching responses in the browser

Answer: C

Q46. [Difficulty: 8] What is the purpose of the TCP TIME_WAIT state?

A) To limit the bandwidth of the connection
B) To wait for the user to type a URL
C) To allow the server to resend its SYN
D) To make sure the final ACK is delivered and old duplicate packets expire

Answer: D

Q47. [Difficulty: 8] What does Nagle's algorithm do?

A) Combines small outgoing segments to reduce the number of tiny packets
B) Encrypts small packets
C) Splits large packets into smaller ones
D) Increases the congestion window faster

Answer: A

Q48. [Difficulty: 8] Which feature does HTTP/2 add compared to HTTP/1.1?

A) Removing headers completely
B) Only allowing the GET method
C) Multiplexing several streams over a single TCP connection
D) Switching from TCP to UDP

Answer: C

Q49. [Difficulty: 8.5] HTTP/3 runs on top of which transport?

A) QUIC over UDP
B) TCP
C) SCTP
D) ICMP

Answer: A

Q50. [Difficulty: 8.5] How does TCP slow start work?

A) The receiver slowly increases its buffer size
B) The sender transmits at full speed and slows down on the first ACK
C) The window grows by exactly one segment per second
D) The congestion window starts small and roughly doubles every round trip until a threshold

Answer: D

Q51. [Difficulty: 8.5] What is the purpose of CORS in web browsers?

A) It lets servers specify which other origins may access their resources
B) It resolves domain names faster
C) It compresses responses
D) It encrypts cookies

Answer: A

Q52. [Difficulty: 9] When does TCP perform a fast retransmit?

A) When the connection is closed
B) After receiving three duplicate ACKs, without waiting for the timeout
C) When the receiver window becomes zero
D) After every 3 seconds

Answer: B

Q53. [Difficulty: 9] What does a SYN flood attack do?

A) Exhausts the server's half-open connection queue by sending SYNs without completing the handshake
B) Steals session cookies
C) Poisons the DNS cache
D) Floods the network with ping packets

Answer: A

Q54. [Difficulty: 9] Which protocol exchanges routing information between autonomous systems on the Internet?

A) ARP
B) RIP
C) OSPF
D) BGP

Answer: D

Q55. [Difficulty: 9.5] Which algorithm does OSPF use to compute routes?

A) Bellman-Ford
B) Dijkstra's shortest path algorithm
C) Floyd-Warshall
D) Prim's algorithm

Answer: B

Q56. [Difficulty: 9.5] What is the count-to-infinity problem?

A) A TCP sequence number overflowing
B) Distance-vector routers slowly increment the cost of an unreachable route; split horizon helps reduce it
C) DNS recursion that never ends
D) A switch learning too many MAC addresses

Answer: B

Q57. [Difficulty: 9.5] How does Path MTU Discovery work?

A) It splits all packets into 512 bytes
B) It asks the DNS server for the MTU
C) It sends packets with the Don't Fragment bit set and reads ICMP 'Fragmentation Needed' replies
D) It uses ARP to learn the MTU of each router

Answer: C

Q58. [Difficulty: 10] What is the 'silly window syndrome' in TCP?

A) A deadlock between SYN and FIN packets
B) The sender window growing without limit
C) Packets arriving out of order
D) Sender and receiver exchange tiny segments because of small window advertisements

Answer: D

Q59. [Difficulty: 10] How many round trips does a full TLS 1.3 handshake need before application data can be sent?

A) 3 round trips
B) 0 round trips
C) 2 round trips
D) 1 round trip

Answer: D

Q60. [Difficulty: 10] What is anycast?

A) One IP address is announced by several servers and traffic goes to the nearest one
B) Sending a packet to a selected group of hosts
C) Sending a packet to every host on the network
D) Sending a packet to one specific host

Answer: A